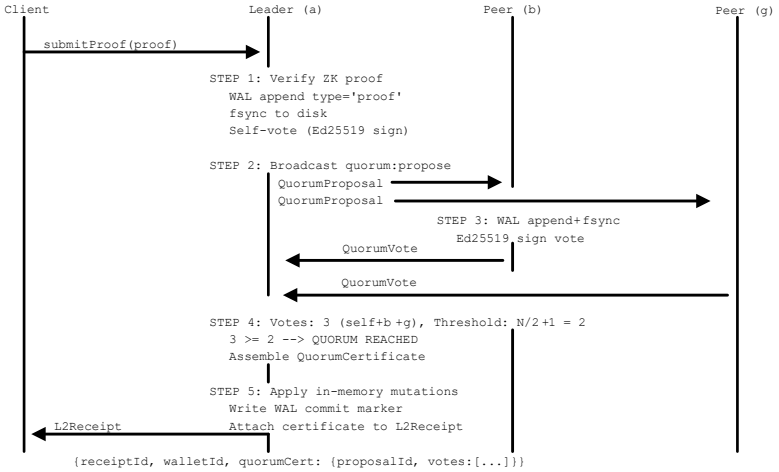
FIG. 15B — QUORUM DURABILITY PROTOCOL, CERTIFICATES & CRASH RECOVERY 1504

SECTION C: QUORUM WRITE PROTOCOL (5-STEP) 1504



TIMEOUT: Peer g DOWN --> votes=2 (self+b), threshold=2, 2>=2 --> QUORUM OK (degraded)
Both peers DOWN --> votes=1, wait 5000ms --> TIMEOUT, reject: "Quorum timeout: 1/2 votes"

SECTION D: CERTIFICATE STRUCTURES 1506

QuorumCertificate (Per-PROOF, on L2Receipt): StateCertificate (Per-BATCH, on L2Batch):

```
proposalId: "a3f2.." (= WAL entryId)
partitionIndex: 0
sequenceNumber: "42"
walEntryHash: SHA-256 of WAL entry
stateDigestHash: hash of post-state
leaderId: "alpha"
createdAt: 1709312001000
votes: [
   {voterId:"alpha", signature:"3a4b..",
    walEntryHash:"7b8c..", ts:...},
   {voterId:"beta", signature:"8f9e..",
    walEntryHash:"9d0a..", ts:...}
]

Proves: N/2 + 1 nodes durably WAL'd this
   proof before client ack
```

Scope: per-PROOF | Freq: every submission
Sign: zkpcare-quorum-vote-v1

```
version: 1
batchNumber: "7", partitionIndex: 0
stateDigest: {
   latestSequenceNumber, latestBatchNumber,
   auditChainHash, claimAccumulatorRoot,
   registryRoot, lifecycleRoot,
   walletCount }
stateDigestHash: SHA-256 pipe-delimited
signatures: [
   {nodeId:"alpha", signature:"3a4b..",
    publicKeyHex:"7f8e.."},
   {nodeId:"beta", signature:"8f9e..",
    publicKeyHex:"2c3d.."}]

Canonical: "zkpcare-state-cert-v1|7|0|
   {hash}|{nodeId}|{ts}"
Proves: N/2 + 1 agree on state at batch
   boundary. Replaces L1 as trust proof.
```

Scope: per-BATCH | Freq: every batch build

SECTION E: CRASH RECOVERY 1508

Node restarts after crash:

```
PASS 1: COLLECT COMMITS
Read WAL line-by-line
if has "commit" field -->
   add entryId to commitSet
(skip corrupt/partial lines)
```

```
PASS 2: LOAD ENTRIES
Read WAL line-by-line
if commit marker --> skip
if entryId in commitSet -->
   entry.committed = true
Store in entries Map
```

```
REPLAY UNCOMMITTED ENTRIES
getUncommitted(partIdx) sorted by sequenceNumber

  type='proof': Apply receiptData, auditEventData, claimEntryData,
     advance roots from receipt's post-mutation values, write commit marker
     NO RE-PROVING NEEDED (all outputs pre-computed in WAL entry)

  type='batch_begin' (no matching batch_commit): DISCARD (incomplete batch)

  type='settle'/'fail': Apply settlement/failure status, write commit marker

Node fully recovered. Resume normal operation.
```